# IT5601
# Embedded Systems and Internet of Things

**Smart City**
**Designing a Level 6 IoT System**

Done By


| | |
|---|---|
| Mohamed Jamal Basha K | 2022506001 |
| Anushri T | 2022506075 |
| Niyathi S B | 2022506077 |
| Yuvaraj K | 2022506079 |
| Manobala P | 2022506081 |
| Nithish S | 2022506099 |
| Krishnaa S | 2022506104 |

**Department of Information Technology**
**Madras Institute of Technology,**
**Anna University, Chennai – 600 044**


January 2025 – May 2025

# Table of Contents


1 What is a Smart City? 1
2 Traffic Management 3
2.1 Intelligent Traffic Lights 3
2.2 Smart Zebra Crossings 4
2.3 Accident Detection and Response 5
3 Public Well-Being 6
3.1 Noise-Free Zones 6
3.2 Pothole Detection and Repair 7
3.3 Criminal Detection and Tracking 8
3.4 Digital Citizen 9
4 Transport 10
4.1 Parking Guidance 10
4.2 Parking Violation Detection 11
4.3 Smart Street Lighting 12
4.4 Smart Toll Systems 13
5 Cleanliness 14
5.1 Intelligent Public Toilet Maintenance 14
5.2 Smart Waste Segregation 15
5.3 Underground Waste Conveyor 16
5.4 Littering Management 17
5.5 Smart Drainage Cleaning 18
6 Infrastructure 19
6.1 Smart Buildings 19
6.2 Smart Billboards 20
6.3 Smart Bus Stands 21
6.4 Rainwater Harvesting 22
6.5 Water Quality Monitoring 23
7 Level 6 Diagram 24

# What is a Smart City?

According to Wikipedia,

*A smart city is an urban area that uses digital technology to collect data and operate services. Data can be collected from citizens, devices, buildings, or cameras.*

Some early definitions of smart cities include:

| | |
|---|---|
| Caragliu et al. (2011) | "A city is smart when investments in human and social capital and traditional (transport) and modern (ICT) communication infrastructure fuel sustainable economic growth and a high quality of life, with a wise management of natural resources, through participatory governance." |
| Bakici, Almirall, & Wareham (2013) | "Smart city as a high-tech intensive and advanced city that connects people, information, and city elements using new technologies in order to create a sustainable, greener city, competitive and innovative commerce, and an increased life quality." |
| Nam and Pardo (2011) | "A smart city infuses information into its physical infrastructure to improve conveniences, facilitate mobility, add efficiencies, conserve energy, improve the quality of air and water, identify problems and fix them quickly, recover rapidly from disasters, collect data to make better decisions, deploy resources effectively, and share data to enable collaboration across entities and domains." |

Basically, just like how a smartphone is said to be a phone that is *smart*, a smart city can be considered as a city that is smart. It achieves this smartness in quite a simple manner of integrating different modes of data collection and performing actions based on what is processed and analysed.

It is essentially the integration of people, technology, and associated processes to connect, interact, and excel across different sectors such as healthcare, transportation, education, infrastructure, and so much more.

Smart city initiatives require three major hallmarks, as previously defined by the U.S. Department of Transportation:

- Network of sensors and citizens to collect data.
- Connectivity from the networks, data, and citizens to the government.
- Open data to share the results, changes, improvements, and data fairly to the public.

The overarching mission of a smart city is to optimise city functions, drive economic growth, and most importantly, improve the quality of life for its citizens using technology and data analysis. How potent and valuable and functional a smart city is based on what it chooses to do with all the technological arsenal that it possesses, not just how much technology it may actually have.

Smart cities integrate Information and Communication Technologies (ICT), devices connected to the Internet of Things network such as sensors and actuators, Artificial Intelligence, and automation of tasks.

The different components of smart cities include, but are not limited to:

1) Traffic and Accident Management
2) Environment
3) Parking
4) Crime and Threat Management
5) Advertising
6) Infrastructure
7) Energy
8) Communication

# Traffic Management

Traffic management is all about regulating and optimising traffic flow using intelligent technologies like adaptive traffic lights and accident detection. Its main focus should be on minimising traffic congestion along particular lanes, improving road safety for vehicles and pedestrians, and reducing travel time.

**Intelligent Traffic Lights**

- Smart traffic lights  that adapt to real-time traffic flow and circumstances.
- Provides higher priority for emergency vehicles such as ambulances.
- Especially beneficial in junctions and central city areas.

| **Device** | **Purpose** |
|---|---|
| Vehicle Counter | To measure vehicle density in each lane |
| Smart Timer | To set signal timer based on traffic |
| RFID Reader | To detect emergency vehicles |
| Smart Controller | To change traffic lights dynamically |
| Alert Node | Periodically sends traffic summary reports |

**Analyses**

- **Emergency Response Optimization:** Analyse the frequency and response time of emergency vehicles to optimize signal timing for faster clearance.
- **Traffic Pattern Recognition:** Study peak and off-peak traffic patterns to dynamically adjust light cycles and reduce congestion.
- **Infrastructure Enhancement Mapping:** Identify consistently congested zones to suggest road expansions or alternate routes.

**Smart Zebra Crossings**

- Detect pedestrians near zebra crossings.
- Accordingly alter the colour of the zebra crossings.
- A stark, bright, distinct red when pedestrians are present, and a similar green when the vehicles are free to go.

| **Device** | **Purpose** |
|---|---|
| Smart Camera | To detect pedestrians |
| Motion Sensor | To detect motion on the crossing |
| Ultrasonic Sensor | To detect incoming vehicles |
| Controllers | Alters colour of the zebra crossing |

**Analyses**

- **Pedestrian Flow Timing:** Analyse the average time pedestrians take to cross to fine-tune signal durations and colour change timings.
- **Peak Crossing Periods:** Identify hours with high pedestrian activity to prioritize alertness and responsiveness during those times.
- **Driver Compliance Monitoring:** Track how often vehicles stop during red zebra signals to assess the need for additional enforcement or awareness.

**Accident Detection and Response**

- Detect accidents and the type of accident that has occurred.
- Notifies the necessary emergency services.
- Send the recorded data to a cloud server.

| **Device** | **Purpose** |
|---|---|
| Accelerometer, Gyroscope | To detect sudden impacts |
| Flame and Smoke Sensor | To detect fire |
| Temperature Sensor | To confirm overheating |
| Location Monitor | Uses GPS to send location information |
| Microphone | To detect sudden, high-decibel noises |
| Alert Node | Uses Wi-Fi for communication |

**Analyses**

- **Accident Frequency Mapping:** Analyse the locations and times of frequent accidents to identify high-risk zones and optimize response readiness.
- **Accident Type Correlation:** Examine the relationship between accident types and environmental or behavioural factors to enhance detection accuracy.
- **Response Time Optimization:** Measure emergency response times to different types of accidents to identify and reduce delays in service deployment.

# Public Well-being

The beating heart and core of any city is its citizens. Ensuring safety and quality of life in an urban environment through intelligent systems is the idea of public well-being. It is implemented through controlling noise pollution, efficient maintenance of roads, and proper detection and handling of different types of crimes. Citizens should also be able to be a part of the data, providing complaints and reports that must be handled accordingly.

**Noise-Free Zones**

- A system that monitors noise pollution in areas such as schools and hospitals.
- Display alerts if the noise level exceeds a prescribed limit.
- Store information about vehicle or whatever it is that caused the pollution.
- Check for repetitive actions and fine accordingly.

| **Device** | **Purpose** |
|---|---|
| Sound Sensor | To measure decibel level |
| Sound Classifier | AI system to differentiate sounds |
| Warning System | To warn polluters through an LED display |
| AI Cameras | To detect polluting actions and polluters |
| Alert Node | To send notifications to the server/police |
| Fine Enforcer | To calculate and send the necessary fine |

**Analyses**

- **Noise Peak Time Mapping:** Track the times of day when noise levels are consistently high to plan preventive measures.
- **Pollution Source Identification:** Log frequent polluters and sound types to inform stricter enforcement policies.
- **Violation Frequency Trends:** Analyse repeated violations in specific locations to determine the need for stricter local regulations.

**Pothole Detection and Repair**

- Detects potholes and similar damages in roads.
- Initiate repair mechanisms using automated bots.
- Perform repairs only when there is no roadside disturbance.
- Keep track of location information for analytics.

| **Device** | **Purpose** |
|---|---|
| Smart Camera | To detect potholes/other damages |
| Ultrasonic Sensor | To detect roadside disturbances |
| Location Monitor | To keep track of damage location |
| Autobots | To provide repairs |
| Alert Node | To notify human professionals |
| Smart Display | To alert incoming traffic of road damage |

**Analyses**

- **Damage Density Mapping:** Analyse which roads have recurring potholes to prioritize repair efforts.
- **Repair Effectiveness Evaluation:** Track how long repairs last before issues recur to assess repair quality.
- **Disturbance-Aware Scheduling:** Check the correlation between traffic density and successful repair timings to avoid delays.

**Criminal Detection and Tracking**

- Identify known criminal(s) when spotted on a public CCTV camera.
- Send the location to the authorities.
- Keep tracking the identified criminal(s) in real-time through nearby cameras.

| **Device** | **Purpose** |
|---|---|
| AI-enabled CCTV | Facial recognition and tracking of criminals |
| Location Monitor | To keep track of criminals' whereabouts |
| Motion Sensors | To detect motion, direction, suspicious activities |
| Alert Node | Sends necessary info to authorities |

**Analyses**

- **Hotspot Activity Analysis:** Identify areas with frequent sightings of known criminals to deploy more resources.
- **Tracking Chain Monitoring:** Measure how efficiently the system hands off tracking between cameras.
- **Suspicious Movement Trends:** Review motion sensor data to improve anomaly detection for non-identified threats.

**Digital Citizen**

- Enables citizens to post complaints on a digital platform.
- The truth of the complaint is verified and the necessary officials are informed.
- Trustworthy user, multiple corroborations are indicators of truth.
- Manual action is then taken.

| **Device** | **Purpose** |
|---|---|
| Capacitive Sensor | For touch-based interaction |
| Microphone | For voice-based reporting |
| Location Monitor | Tags the location of the complaints |
| Smart Display | Shows related complaints and reports |
| Speaker | Voice-based notifications or feedback |
| Alert Node | Forwards high-priority complaints |

**Analyses**

- **Complaint Clustering:** Analyse complaints by location and type to identify priority areas needing intervention.
- **User Credibility Evaluation:** Study patterns in user reliability to auto-prioritize genuine complaints.
- **Response Time Monitoring:** Evaluate how quickly each complaint is addressed to improve service responsiveness.
- **Issue Hotspot Mapping:** Identify geographic locations with high complaint frequency to prioritize attention and resource allocation.

# Transport

There are various means of transport in an urban environment. There are of course various private modes of transport through two-wheelers and four-wheelers, but also public transport through autos, buses, and trains. Automating various factors in these areas to improve the satisfaction of commuters goes a long way in making a city smart in its transport sector.

**Parking Guidance**

- Parking lots that indicate where a vehicle should park through smart displays .
- Analyses already filled spots, detects the type of vehicle, and accordingly provides directions.
- Communication between multiple smart displays.

| **Device** | **Purpose** |
|---|---|
| Smart Mapper | Maps the entire parking space, showcasing empty parking spots and filled spots |
| Ultrasonic Sensor | Equipped at each parking spot to send status info |
| Smart Camera | Detects incoming vehicles, type of vehicle |
| Counter Display | Shows total number of parking spots, no. of occupied spots, and no. of free spots in real-time |
| Smart Gate | Placed at different sections, closes and opens based on availability of parking spots |
| Smart Display | To direct vehicles to parking spots |

**Analyses**

- **Occupancy Trend Analysis:** Track time-based usage patterns to predict peak hours and optimize parking space availability.
- **Vehicle-Type Allocation Efficiency:** Analyze how accurately vehicle types are matched with designated spots to improve space utilization.
- **Display Communication Accuracy:** Monitor consistency between multiple smart displays to ensure synchronized and error-free directions.

**Parking Violation Detection**

- Monitors no-parking zones using smart cameras and/or specialized readers.
- Detects illegal parking, captures evidence.
- Notifies the respect traffic authorities immediately.

| **Device** | **Purpose** |
|---|---|
| Ultrasonic Sensor | To detect presence of vehicles in no-parking zones |
| Smart Camera | To detect individuals who made the parking violation |
| RFID Reader | To read license plates and recognize vehicles |
| Display | To notify drivers of their parking violation |
| Location Monitor | Marks location of parking violation |
| Alert Node | Sends information to respective authorities |

**Analyses**

- **Violation Frequency Analysis:** Analyze when and where violations occur most frequently to improve surveillance and enforcement planning.
- **Offender Recurrence Tracking:** Track repeated offenses by the same vehicles to identify habitual violators and take stricter action.
- **Alert Response Monitoring:** Measure how quickly authorities respond to alerts to enhance system efficiency and accountability.

**Smart Street Lighting**

- Street Lights that are automatically turned on and off based on outdoor conditions.
- More efficient and effective than being turned on and off manually or based on some timer.
- Lighting conditions, vehicle/pedestrian density, are all considered.

| **Device** | **Purpose** |
|---|---|
| Light Sensor | To measure the intensity of light |
| Motion Sensor | To detect vehicles/pedestrian motion |
| Weather Sensor | To detect rain, fog, etc. |
| Real-Time Clock | Default settings of on and off |
| Smart Light | Controls brightness level |
| Alert Node | Sends lighting information for a specific region |

**Analyses**

- **Lighting Optimization:** Analyse how the intensity of street lighting can be adjusted based on ambient light levels to save energy while maintaining safety.
- **Traffic Flow Analysis:** Study the vehicle and pedestrian density data to optimize the timing of street light activation, ensuring adequate illumination when traffic is highest.
- **Maintenance Prediction:** Use system data to predict when individual lights may need maintenance or replacement based on usage patterns and performance metrics.

**Smart Toll Systems**

- Automated toll systems with smart cameras that detect every vehicle passing through.
- The FastTag card or the number plate is scanned and accordingly charged.
- The long queues and manual workload are avoided.

| **Device** | **Purpose** |
|---|---|
| RFID Reader | To read FastTag card |
| Smart Camera | To read number plate through OCR |
| Motion Sensor | To specifically detect vehicles |
| Speed Sensor | To measure speed of vehicles |
| Activator | Activates cameras and readers upon arrival of vehicles |
| Alert Node | Updates payment charge on vehicles |

**Analyses**

**Optimize Toll Efficiency:** Analyse peak hours and vehicle patterns to optimize lane distribution and toll system response time.
**Minimize Revenue Leakage:** Identify trends in non-payment incidents by examining FastTag failures or unregistered vehicles to improve compliance measures.
**Enhance System Reliability:** Monitor system performance data to predict and prevent failures in cameras and scanners for uninterrupted operation.

# Cleanliness

In any city, however modernised and advanced it may be, there will always be litter and waste just flying all over. With such powerful technology and structures, it baffles the mind to see such poor management of waste. In a smart city, waste management is automated, and any obstructions to cleanliness in the public is duly punished through fines, in order to promote and ensure a proper livelihood, keeping cities clean and green.

**Intelligent Public Toilet Maintenance**

- Public toilets that clean and maintain themselves on a regular basis.
- Intelligent measures put in place to punish those who deliberately vandalise the toilets.

| **Device** | **Purpose** |
|---|---|
| Odour Sensor | To detect foul odour |
| Damage Sensor | To detect damaged doors/urinals/toilets/flush |
| Water Sensor | To detect water overflow |
| Dirt Sensor | To detect unnecessary dirt and untidiness |
| Smart Cameras | To recognise vandals (present outside) |
| Autobot | Makes necessary repairs |
| Alert Node | Alerts officials for repairs |

**Analyses**

- **Usage Patterns Analysis:** To optimise cleaning schedules and maintenance routines.
- **Vandalism Trends Identification:** Identify patterns in vandalism occurrences to implement targeted deterrent measures.
- **Operational Efficiency Evaluation:** Assess cleaning and maintenance performance metrics to identify inefficiencies and recommend improvements in resource allocation or system design.

**Smart Waste Segregation**

- Throw waste into a singular public dustbin.
- It automatically segregates and separates the waste.
- When the dustbins reach a tipping point, officials are informed to collect and handle the waste.

| **Device** | **Purpose** |
|---|---|
| Smart Waste Sensor | Detects type of waste for segregation |
| Fill-Level Sensor | To measure level of waste in dustbin |
| Smart Compressor | Compresses waste for maximum bin utilisation |
| Smart Divider | Segregates the waste based on sensor data |
| Alert Node | Sends an alert when dustbins are nearly full |
| Display | Displays percentage of bin filled |

**Analyses**

- **Waste Segregation Efficiency:** Analyse the percentage of correctly segregated waste to identify areas where the system may need improvement.
- **Collection Optimization:** Track the fill levels of dustbins to optimize collection routes and schedules, reducing unnecessary trips.
- **Public Engagement Insights:** Monitor the frequency of usage patterns to design better awareness campaigns or improve public participation in waste segregation.

**Underground Waste Conveyor**

- Vacuum-powered underground pipeline system for waste transport.
- Sucks waste from nearby smart bins.
- Transfers it to a centralized unit, wherein recycling, composting, etc. take place.

| **Device** | **Purpose** |
|---|---|
| Master Sensor | Gets fill-level data from all surrounding dustbins |
| Integrity Sensor | Monitors structural integrity |
| Leak Detection Sensor | Monitors leaks in pipeline system |
| Vacuum System | Pulls waste and regulates flow |
| Pipeline Actuator | To direct waste flow based on section availability |
| Conveyor Belt | Transports waste |
| Alert Node | Sends maintenance alerts to officials |

**Analyses**

- **System Efficiency Monitoring:** Analyse the vacuum power usage to assess the system's energy efficiency and identify areas for improvement.
- **Waste Flow Optimization:** Track the speed and volume of waste transfer to ensure the pipeline is operating at optimal capacity.
- **Maintenance Prediction:** Monitor wear and tear data to predict maintenance needs, minimizing downtime and preventing system failures.

**Littering Management**

- Detect littering in public places.
- Identify the responsible individuals through their face and Aadhaar ID.
- Automatically send a fine.
- Analyse frequently littered areas.

| **Device** | **Purpose** |
|---|---|
| AI-Enabled Camera | Real-time video analysis and facial recognition |
| Location Monitor | Uses GPS to map the location of the littered zone |
| Motion Sensor | To detect activity in frequently littered zones |
| Environmental Sensor | To measure waste-related pollution levels. |
| Fine Enforcer | To calculate and send the necessary fine |
| Buzzer/Speaker | On-spot warnings |

**Analyses**

- **Littering Hotspot Identification:** Analyse data on frequently littered areas to target high-risk locations for more efficient cleaning and enforcement.
- **Fine Effectiveness Assessment:** Track the impact of fines on littering behaviour by analysing changes in littering frequency before and after penalties.
- **Facial Recognition Accuracy:** Evaluate the accuracy of the facial recognition system to ensure reliable identification and minimize false positives.

**Smart Drainage Cleaning**

- Monitor drainage, detect blockages, corrosion, odour issues.
- Take appropriate action to fix the drainage.
- Notify required professionals if the same area is repeatedly affected.

| **Device** | **Purpose** |
|---|---|
| Flow Sensor | To detect changes/blocks in drainage flow |
| Gas Sensor | To detect methane, hydrogen sulphide |
| Odour Sensor | To detect specific changes in odour |
| Water Jet | To clean through heavy gushes of water |
| Cleaner | To clean through a telerobotic/AI-based arm |
| Alert Node | To alert human professionals |

**Analyses**

- **Frequent Blockage Mapping:** Identify areas that face repeated blockages to prioritize maintenance and infrastructure upgrades.
- **Corrosion Pattern Detection:** Analyse sensor data over time to detect early signs of corrosion and schedule timely replacements.
- **Odour Intensity Trends:** Monitor variations in odour levels to predict possible health hazards and automate ventilation or cleaning responses.

# Infrastructure

The first thing that instinctively comes to mind whenever we think about a city or a smart city is how it looks – its infrastructure. In a smart city, traditional infrastructure is transformed with through the automation of certain tasks with an effective usage of different technologies. Smart infrastructure provides data-driven insights and monitoring that enables a robust, sustainable, and future-ready urban landscape.

**Smart Buildings**

- Buildings that automatically measure their structural integrity.
- Automated repairs wherever possible.
- Notifications sent to officials for tasks that lie beyond automation.
- Also detect disasters such as earthquakes.

| **Device** | **Purpose** |
|---|---|
| Integrity Sensor | Measures strain, vibrations, tilts, cracks |
| Disaster Sensor | Seismic, wind, and water leakage |
| Environmental Sensor | Measures temperature and humidity |
| Smart Display | To display warnings |
| Alarm | Sounds in the event of some disaster |
| Autobots | Designed to automate some repairs |

**Analyses**

- **Structural Weakness Tracking:** Analyse historical sensor data to identify patterns of weakening in specific building sections.
- **Repair Efficiency Evaluation:** Monitor the frequency and success rate of automated repairs to optimize maintenance protocols.
- **Disaster Response Optimization:** Study sensor-triggered alerts during disasters to improve early warning accuracy and evacuation planning.

**Smart Billboards**

- Advertisement boards that don't need to be manually replaced every time.
- Based on the occasion and the environment, automatically provides a suitable advertisement.
- Updated remotely with new ads that the billboard itself decides when best to utilise.

| **Device** | **Purpose** |
|---|---|
| Smart Camera | To analyse the demographic present in an area |
| Weather Detector | To provide information based on weather |
| Ultrasonic Sensor | To play animated advertisements when people get nearby; for smaller billboards |
| Light Sensor | Adjust brightness based on outdoor lighting |
| Smart Display | Changes content based on requirements |
| Speaker | For any advertisements with sounds |

**Analyses**

- **Audience Engagement Analysis:** Track viewer interaction and response times to determine which ads capture the most attention.
- **Contextual Ad Performance:** Analyse how different environmental factors affect the effectiveness of specific advertisements.
- **Optimal Display Timing:** Study time-based trends to schedule ads when they are most likely to reach the target audience.

**Smart Bus Stands**

- Bus stands – small or large – as closed smart shelters.
- Provides travel information.
- Security and connectivity are ensured.
- Automatic temperature-based air conditioning.

| **Device** | **Purpose** |
|---|---|
| Environmental Sensor | Monitors temperature, humidity, lighting |
| Occupancy Sensor | Counts no. of people inside |
| Smart Display | Provides personalised travel information |
| Smart Camera | Monitors infractions, security issues |
| Smart AC | AC based on environment and occupancy |
| Alarm | Sounds in the event of some emergency issue |

**Analyses**

- **Occupancy Trends**: Analyse peak and off-peak usage times to optimise space and services.
- **Climate Response Efficiency**: Evaluate how effectively the AC adapts to temperature and crowd levels.
- **Security Pattern Analysis:** Review camera and alarm data to identify frequent security issues or blind spots.

**Rainwater Harvesting**

- Automates rainwater collection, filtration, and quality monitoring.
- Ensures safe storage of quality water in real-time.

| **Device** | **Purpose** |
|---|---|
| Rain Sensor | To initiate harvesting process |
| Water Level Sensor | To monitor water level status in tank |
| Water Quality Sensor | To monitor pH, turbidity, etc. |
| Motorised Valve | Directs water flow to pipeline |
| Filtration Actuator | Performs necessary filtration |
| Storage Actuator | Redirects excess water to secondary storage |

**Analyses**

- **Rain Pattern Insights:** Analysing rain sensor data helps optimise system activation times and storage planning.
- **Tank Utilization Trends:** Tracking water level sensor data reveals storage efficiency and potential overflow risks.
- **Water Quality Monitoring:** Studying quality sensor readings enables early detection of contamination and timely filtration adjustments.

**Water Quality Monitoring**

- Monitors the quality of water delivered through metro pipelines.
- Detects leakage and implements precautionary measures.
- Sends real-time alerts to the required officials wherever necessary.

| **Device** | **Purpose** |
|---|---|
| Water Quality Sensor | To monitor pH, turbidity, etc. |
| Leak Detection Sensor | To monitor pressure, flow, acoustic irregularities |
| Integrity Sensor | Measures strain, vibrations, tilts, cracks |
| Motorised Valve | Shuts off or redirects water flow |
| Pump Actuator | Adjusts water pressure |
| Disinfection System | Neutralises contaminants |
| Alert Nodes | Sends alerts to necessary officials |

**Analyses**

- **Leak Pattern Detection:** Analysing leak detection sensor data highlights pressure drops or flow anomalies, enabling fast pipe repair and reduced water loss.
- **Structural Health Monitoring:** Tracking integrity sensor metrics helps identify pipe strain, vibrations, or cracks, preventing infrastructure failure.
- **Contamination Alerts:** Monitoring water quality sensors ensures pH and turbidity levels stay within safe ranges, triggering disinfection or alerts when needed.